360° RAYS - Radial Visualisation of High Dimensional Data

Abstract. Visualising high-dimensional real data with all dimensions while providing some semantics is a challenging problem in data analysis.
In this paper, we present the visualisation of high-dimensional real data using our implementation tool, ***360° RAYS***. It is a visualisation built to plot high-dimensional data without the loss of information, unlike dimensionality reduction techniques. RAYS helps plot all the features and points in the dataset and aims to provide information such as the **geometrical positioning of points or classes in the orthants of space**, also aiding in understanding the **closeness of multiple points, clustering of points, and ordering of subspaces**. This visualisation is a novel technique built after exploring major viable options to plot higher-dimensional data, which included parallel coordinates, circle segments, among others. The key idea of our approach is a **hierarchical concentric ring structure**, where each successive ring incrementally adds a dimension to the existing subspace, ordered by the variance of data in the added dimension. Data points are mapped to ring sectors based on a **bitwise encoding of their coordinate values** (non-negative or negative).
The advantage of our approach is that it presents the complete data with all the dimensions and provides a **clear visual grouping of points by subspace concentration**. The visualisation also shows classes within the data. Hence, it enables the complete representation of the data in a flattened 2D radial layout.

Keywords: High Dimensional Data · Data Visualization · Radial Geometry · Subspace Analysis · Variance Ordering · Bitwise Encoding

1 Introduction

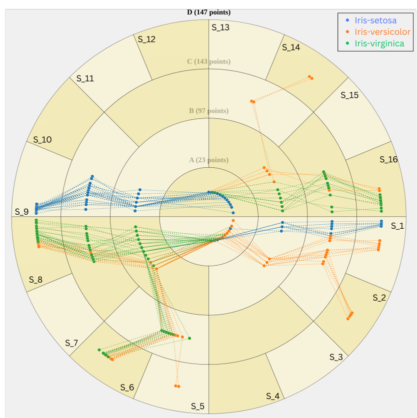
In this research paper, we study various types of data visualisation, such as **parallel coordinates** [1] and **circle segments** [2], that are used to visualise multi-dimensional data, and explain why our visualisation is different and useful in terms of visualising high-dimensional data.

The core idea of **RAYS** is to **visualise high-dimensional data while providing meaningful inferences** to the user and studying this usage of orthants (a geometric term for a multidimensional generalization of a quadrant (2D) or octant (3D)) in space. The visualisation aims to provide a **clutter-free view** of the $\mathbb{R}^d$ data, where d is the dimension of the dataset, along with a meaningful distribution of the data points and the classes (if any).

The goal of the visualisation is to help the user understand the **relative positioning** of the data points and the **usage of different orthants** in the space. This novel technique aims to provide users, typically data scientists, with an overview of the spread and geometric positions of the data points in the dataset. It makes use of the data in its raw form, i.e., without any dimensionality reduction techniques applied, providing a clearer view of the relationship between different points and orthants without the loss of any information.

We will first study how this visualisation is constructed and its structure, and how this structure serves the purposes mentioned above. While discussing the construction, we will also highlight the relevance of the visualisation in the present time.

2 RAYS



A	SepalWidthCm
B	PetalWidthCm
C	SepalLengthCm
D	PetalLengthCm

Fig. 1: Iris Dataset with 3 labels - RAYS Visualisation along with the corresponding class label legend.

RAYS is used to visualise high-dimensional data without the loss of information, essentially without incorporating any dimensionality reduction techniques. This novel visualisation makes use of subspaces and quadrants in which the point can lie, which helps the users study the patterns of the arrangement of points and potentially identify similarities or dissimilarities between the points.

The visualisation includes various different components which, when put together, build the complete visualisation. The visualisation aims to represent a high-dimensional real dataset as a collection of concentric rings divided into sectors (orthants for 2^d space). This structure is fundamentally hierarchical and radial, which is crucial for managing the complexity and preventing the visualisation from becoming congested and cluttered. The data points are distributed across the rings and sectors, and their relationships are shown using edges.

2.1 Our Visualisation

This section introduces the core concepts and terminologies essential for understanding the construction and interpretation of the RAYS visualisation.

1. Dataset Formalisation: We introduce formal definitions of the data structure being analysed:

1. We consider an $\mathbb{R}^d$ real dataset, where $\mathbb{R}$ represents the set of all real numbers and d denotes the **dimensionality of the feature space**.
2. Let the dataset comprise n distinct data points, denoted as $x_1, x_2, x_3, \ldots, x_n$, where each point x_i belongs to the $\mathbb{R}^d$ space.
3. Each data point x_i is represented by a vector of d coordinates: $(x_{i1}, x_{i2}, \ldots, x_{id})$ where $x_{ij} \in [-1, 1]$ for all $i \in \{1, \ldots, n\}$ and $j \in \{1, \ldots, d\}$.
4. Let $a_1, a_2, \ldots, a_d$ be the dimensions or attributes in the dataset.

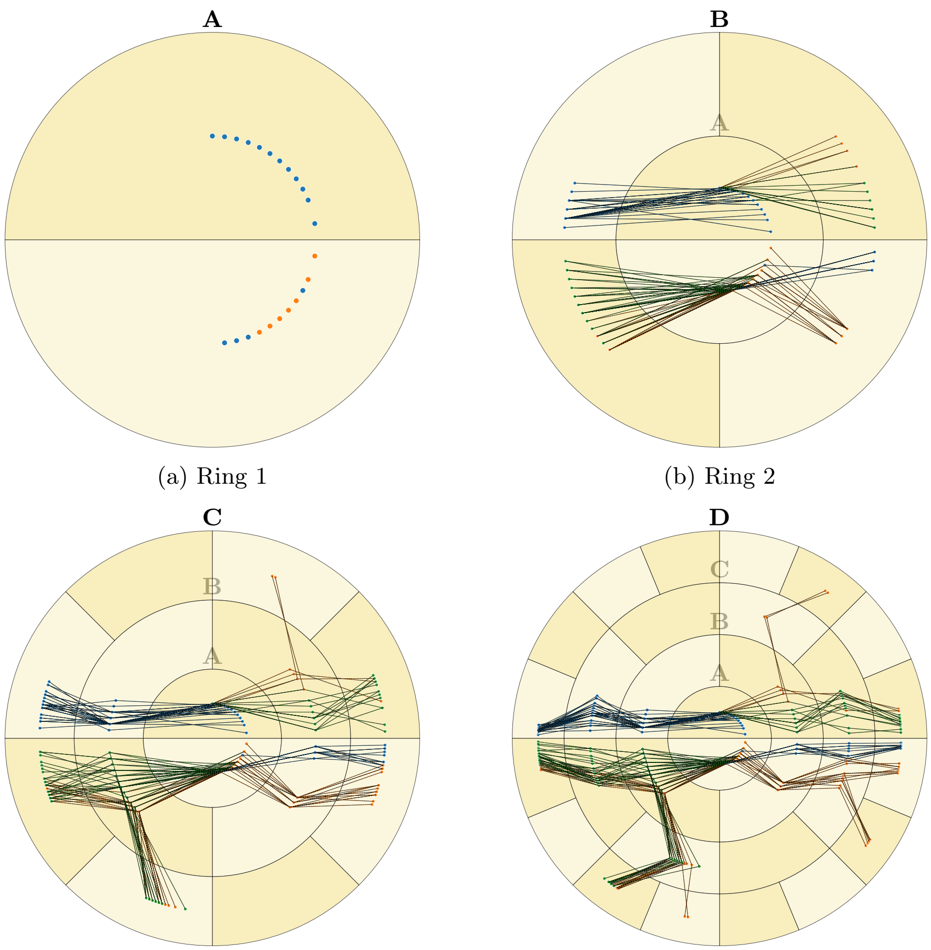
2. Dimension Ordering by Variance: Without loss of generality, we **order the dimensions** $(a_1, a_2, \ldots, a_d)$ based on the **increasing variance**.

Let $\sigma_{a_1}, \sigma_{a_2}, \ldots, \sigma_{a_d}$ be the variance of the d dimensions, and

$$\sigma_{a_1} \leq \sigma_{a_2} \leq \cdots \leq \sigma_{a_d}$$

This ensures that the dimensions contributing the **least** spread (variance) to the data are placed in the innermost rings, providing a stable, low-variance base for subsequent dimensions. Conversely, the dimensions contributing the **most** spread (variance) to the data are strategically placed towards the outer regions. This placement is important because, as observed in the radial geometry, the **area available for partitioning exponentially increases** with the radius. Placing high-variance dimensions in the outer rings provides a larger angular and radial area to contain the greater spread of points, thereby achieving a clearer distribution and minimizing visual clutter and **overlap** during the iterative sector subdivision process. This reduction in clutter with the help of variance can also be learnt from [3], where the author uses different techniques like pattern based clustering to reduce clutter and maximise the pattern identification in parallel coordinates.

3. Hierarchical Concentric Ring Structure The RAYS visualisation is built based upon a hierarchical concentric ring structure. Each successive ring, moving outward from the centre, corresponds to the incremental addition of one dimension, following the variance-based order obtained in the previous step. The rings are labelled A,B,C . . . , as in Figure 2 to represent the subspaces/dimensions.



(a) Ring 1 (b) Ring 2

(c) Ring 3 (d) Ring 4

Fig. 2: Iterative ring construction and point placement (Iris Dataset)

- **Innermost Circle (Dimension a_1):** The innermost circle plots all points' first coordinate, $x_{i,1}$ (where $1 \leq i \leq n$). This circle is fundamentally divided into two major sectors (semi-circles): one for the positive/non-negative values ($x_{i,1} \geq 0$) and one for the negative values ($x_{i,1} < 0$), as shown in Figure 2a. The darker region represents the positive/non-negative sector and the lighter region represents the negative sector. Note: that the color of the rings does not signify any meaning.
- **Successive Rings (Dimension a_j):** The next concentric ring, representing the second dimension a_2, plots the coordinate $x_{i,2}$. The location of $x_{i,2}$ is **relative** to the sector established by $x_{i,1}$, and is further subdivided into two

new sub-sectors based on the sign (positive or negative) of $x_{i,2}$ as shown in Figure 2.

– **Recursive Partitioning and Orthant Mapping:** This process is repeated for every dimension $j = 1$ through d. Each ring j corresponds to the dimension a_j and recursively partitions the angular sectors established by the previous $j-1$ dimensions, based on the sign of the j^{th} coordinate $(x_{i,j})$. After d dimensions, the visualization is segmented into a total of 2^d distinct angular sectors in the outermost ring, which correspond precisely to the 2^d orthants in the $\mathbb{R}^d$ space. The final visualization is obtained after plotting the points in d rings (e.g., Figure 2d). The points' specific angular positions are determined within their assigned sector based on their coordinate value $x_{i,j}$. This position is calculated by:

- **Angular Mapping:** Since the point's coordinate value $x_{i,j}$ lies in the range $[0, 1]$ for positive values, and $[-1, 0]$ for negative values, the visualization maps this value onto the specific angular range of the sector it belongs to. The angular position is determined by treating the sector's arc as a proportional scale: the starting angle represents the minimum possible value of that scale (0 for a positive sector or -1 for a negative sector), and the ending angle represents the maximum value (1 for a positive sector or 0 for a negative sector, proceeding clockwise), allowing the value $x_{i,j}$ to proportionally determine the point's precise angular position along the arc (as shown in Figure 3).

The colors of the points represent class labels (e.g., Figure 1).

4. Point Representation A point's location, up to the $j-1$ dimension, $(x_{i,1}, \ldots, x_{i,j-1})$, is represented by a path that traverses from the sector corresponding to $x_{i,1}$, to $x_{i,2}$, and so on, up to $x_{i,j-1}$ across the $j-1$ dimensions (rings) as shown in Figure 2.

Once the j^{th} dimension is added, the path from $x_{i,j-1}$ to $x_{i,j}$ proceeds based on the value (positive or negative) of the j^{th} coordinate.

The final representation of the complete d-dimensional point $x_i = (x_{i,1}, \ldots, x_{i,d})$ is the **entire continuous path** extending from the innermost concentric circle (Dimension a_1) to the outermost concentric ring (Dimension a_d), as illustrated in Figure 1. This path's angular extent uniquely identifies **orthant** to which the point x_i belongs in the $\mathbb{R}^d$ space.

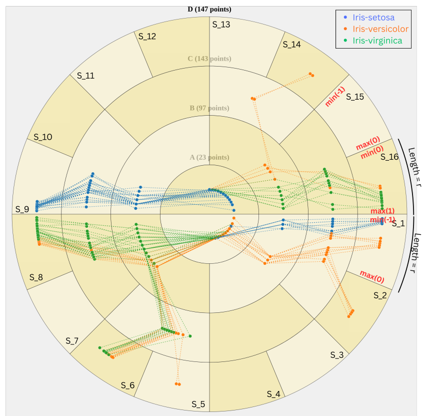


Fig. 3: Iris Dataset points plotted in hierarchical rings and sectors with marking of the scale used to plot the points in the sectors based on their values.

2.2 Algorithm

Using the terminology defined previously, we apply the following algorithm to set up the **RAYS** visualisation structure:

1. **Input:** $\mathbf{X}$, a d-dimensional dataset with n points, $\mathbf{X} = \{x_1, x_2, \ldots, x_n\}$.
2. **Point Format:** Each point x_i is represented as a vector: $(x_{i,1}, x_{i,2}, \ldots, x_{i,d})$ where, $x_{i,j} \in [-1, 1]$ for $1 \leq i \leq n$ and $1 \leq j \leq d$ (after normalisation).
3. **Dimension Ordering:** The indices $j = 1, \ldots, d$ correspond to the dimensions $a_1, a_2, \ldots, a_d$, which are ordered by non-decreasing variance:

$$\sigma_{a_1} \leq \sigma_{a_2} \leq \cdots \leq \sigma_{a_d}$$

Algorithm 1 RAYS Visualisation Construction

procedure ConstructRays($\mathbf{X}$)
 Define the radius r_j and width W_j for each ring $j = 1 \ldots d$.
 Ring Width Definition: The radius of ring j is r_j and its width is $W_j = r_j - r_{j-1}$.
  • r_0 is defined as the center (0). And, $W_j = W$ (constant width).
 Initialisation (Dimension a_1):
 Create the innermost concentric circle (Ring 1) with radius r_1.
 Partition the circle into two sectors based on the sign of $x_{i,1}$:
  • One sector $S_{1,+}$ for $x_{i,1} \geq 0$ (e.g., the non-negative half).
  • One sector $S_{1,-}$ for $x_{i,1} < 0$ (e.g., negative half).
 Place all points' first coordinates $x_{i,1}$ into the corresponding sectors of the innermost circle.
 The angular position $\Theta_{i,1}$ for point i is calculated within $S_{1,+}$ or $S_{1,-}$.
 for $j = 2$ **to** d **do**
  Iterative Partitioning (Dimension a_j):
  Create the j^{th} concentric ring (radius r_j, width $W_j = W$).
  for each existing path segment $P_{i,j-1}$ ending at dimension a_{j-1} in ring $j-1$ **do**
   Let $S_{i,j-1}$ be the sector containing $P_{i,j-1}$.
   Subdivide the corresponding sector in ring j into two new sub-sectors $S_{i,j,+}$ and $S_{i,j,-}$.
   The subdivision is based on the sign of the current coordinate $x_{i,j}$:
    • $S_{i,j,+}$ for $x_{i,j} \geq 0$.
    • $S_{i,j,-}$ for $x_{i,j} < 0$.
   Point Placement and Path Extension:
   Calculate the angular position $\Theta_{i,j}$ for point i by scaling the value $x_{i,j}$ to the angular range of the determined sub-sector $S_{i,j}$.
   Place the point $x_{i,j}$ at the polar coordinates $(r_j, \Theta_{i,j})$.
   Extend the path segment $P_{i,j-1}$ to the new point position at $(r_j, \Theta_{i,j})$ in $S_{i,j}$.
 Output:A radial layout where each point x_i is represented by a unique continuous path from r_1 to r_d, identifying its orthant/sector based on the bitwise sign encoding of its coordinates $(x_{i,1}, \ldots, x_{i,d})$. The points are located in its orthant/sector based on its value relative to its previous dimension's point position as shown in Figure 3.

The RAYS algorithm can be modified to enhance visual intuitiveness. For example, in Figure 4, a modified version of the algorithm is implemented to **collapse any empty sectors**, thereby maximizing the angular space allocated to occupied regions. Concurrently, the angular width of the occupied sectors is made **proportional to the number of data points** they contain. This modification provides the user with an immediate overview of the **point distribution** across the high-dimensional space without requiring detailed point-by-point inspection.

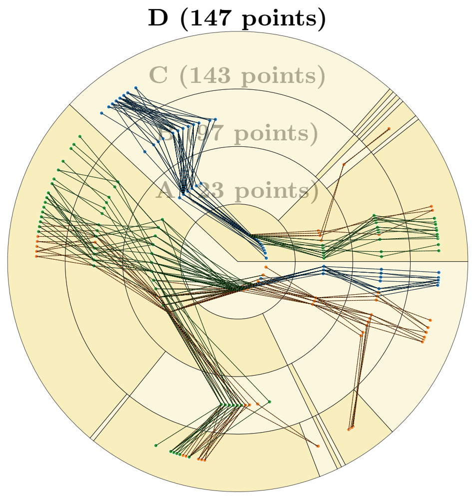
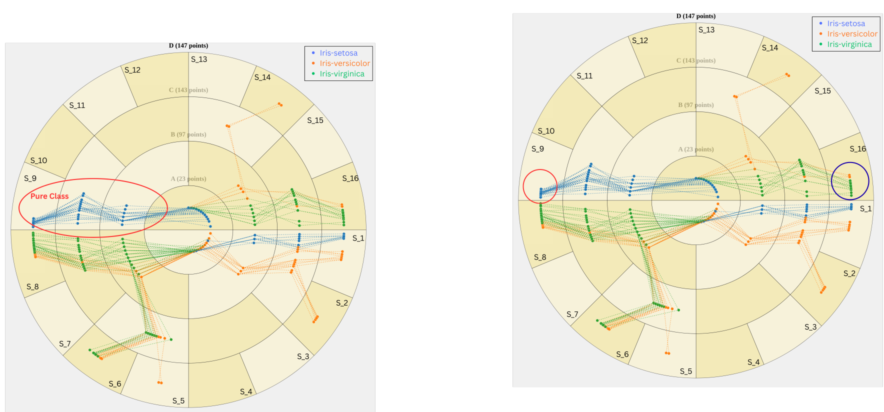


Fig. 4: Iris Dataset visualised in hierarchical rings and sectors using modified algorithm.

2.3 Properties



(a) Pattern Identification (Iris-Setosa)

(b) Red Marking - Pure class; Blue Marking - Mixed Class

Fig. 5: Iris Dataset - RAYS Visualisation

The **RAYS** visualisation exhibits the following key properties, which significantly aid the user in identifying underlying patterns and structural characteristics within the dataset:

t_1: **Spread and Sparsity (Orthant Usage):** As the dimensionality d increases, the $\mathbb{R}^d$ space becomes sparse because the number of orthants, 2^d, grows exponentially faster than the number of typical data points (n). The RAYS visualization explicitly depicts the **usability of the sectors/orthants**, allowing analysts to understand the **spread** and geometric concentration of the data across the high-dimensional space.

t_2: **Continuity and Clutter Reduction:** The visualization leverages the **hierarchical nature of the sectors**, where the angular region of a parent sector fully includes its children. This structure guarantees that the paths representing individual points maintain **continuity**. Crucially, the edges connecting successive coordinates of a single point are assured **never to criss-cross** the edges of any other point within the sector linkages. This feature minimizes visual clutter, ensuring that each point x_i is clearly represented by a **unique continuous path** from the innermost circle (r_1) to the outermost ring (r_d), for $1 \leq i \leq n$.

t_3: **Commonality and Grouping:** If the input dataset $\mathbf{X}$ is a classified dataset (i.e., has class labels), RAYS offers a method to analyse the **commonality and purity of sectors/orthants**. For example: if a sector S_j within a specific ring R_i contains point coordinates belonging exclusively to one class, we define that sector as **pure** (as in Figure 5b, sectors like S_2, S_5, S_9, S_{14} in the outermost ring containing only one type of points). Conversely, if the sector contains a mixture of various classes, we define it as **mixed**. This analysis allows the user to quantify how many sectors are pure versus mixed as shown in Figure 5, providing a data analyst with insights into the general **geometric distribution of classes** in the dataset.

3 Complexity Analysis

The complexity of RAYS visualization setup is analyzed based on the number of data points (n) and the dimensionality of the dataset (d).

RAYS visualization process iterates through all n data points, and for each point, it processes all d dimensions to map the point's path onto the radial structure using the bitwise sign encoding. The core operation for each coordinate is $\mathcal{O}(1)$. Therefore, the total number of operations is proportional to the total number of coordinates in the dataset, $n \times d$.

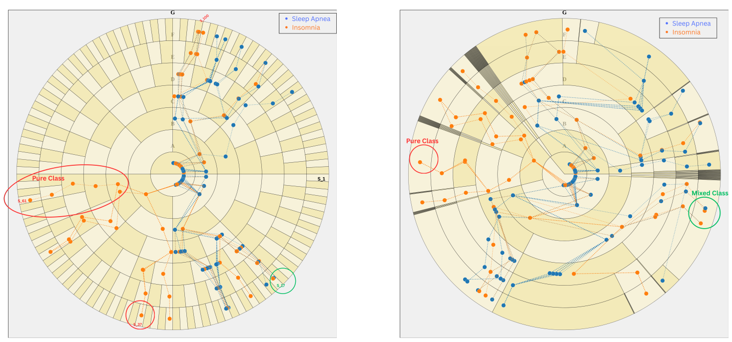
$$\text{Time Complexity} = \mathbf{O}(\mathbf{nd})$$

The space requirement is determined by storing the input dataset $\mathbf{X}$ (an $n \times d$ matrix) and the resulting RAYS visualization structure. The structure stores n paths, each containing d segments, which also requires memory proportional to $n \times d$. Since both the time and dominant space requirements are proportional to the total size of the input data, the overall space complexity is also:

$$\text{Space Complexity} = \mathbf{O(nd)}$$

4 Visual Aspects

The **RAYS** visualisation is designed to provide specific visual cues that facilitate pattern recognition and data interpretation:



(a) (b)

A	SepalWidthCm
B	PetalWidthCm
C	SepalLengthCm
D	PetalLengthCm

(c)

Fig. 6: Sleep-Lifestyle (7-dimensional) Dataset - RAYS Visualisation comparing (a) uniform sector partitioning and (b) proportional sector partitioning, with (c) the ring label legend.

- **Orthant Usage:** The visualisation directly facilitates the identification of highly used sectors versus those that remain empty (as mentioned in property t_1), meaning no data point belongs to the corresponding high-dimensional orthant in the $\mathbb{R}^d$ space as in Figure 6a, we observe the high number of empty orthants/sectors. Due to the sparsity of high-dimensional data, many such unused sectors can be readily identified. This capability provides advantage over techniques like **Circle Segments** [2], which primarily provide a density-based overview using pixel-based representations, often obscuring the underlying geometric structure. We can also remove the empty sectors as in Figure 6b, to get on overview of the distribution at a glance.

- **Point Identification and Cognitive Load Reduction:** Identifying and tracing a specific data point based on its position is simplified due to the **hierarchical nature of the visualisation**, as elaborated in property t_2. Unlike **Parallel Coordinates** [3], where understanding a complete data point requires visually tracing a line across d separate axes and remembering prior information of the data point (increasing cognitive load), RAYS represents the point as a single, continuous, angular path across rings.This hierarchical structure reduces the cognitive load by storing the information of previous coordinates within the current sector's position. Furthermore, the bitwise encoding embedded in each sector allows the user to identify the point's orthant (sign pattern) at any given ring without needing to recall prior coordinate details.

- **Integrated Class Semantics:** RAYS effectively integrates class information directly into the visual structure. By using distinct colours or visual properties for different classes, the visualisation reveals the geometric positions of these classes within the defined sectors/orthants. As depicted in Figure 6b, we can observe that some sectors are pure sectors (marked in red) and some sectors are mixed sectors (marked in green) containing mixed classes. In Figure 6a, we can also observe that the paths of the points in sector S_{61} are all orange, signifying that the parent sectors of S_{61} are also pure. Track of purity of classes can be made through RAYS and this is important, just like in [4], the authors signifies that there is often a changing pattern due to the impact of attributes.

5 Datasets and Analysis

Public Repository and Results: All datasets used for testing and validation, along with the corresponding output visualisations and detailed analytical results, are made publicly available.The resources can be accessed at the public repository TO BE UPDATED

6 Conclusion

The visualization of high-dimensional data remains a significant challenge in data analysis. While a large number of techniques exist to plot such data, the **RAYS** visualization presents a novel solution by directly mapping the **raw, high-dimensional data** into a flattened 2D radial layout, thereby completely preserving the original information without relying on dimensionality reduction.

The primary contribution of RAYS lies in its ability to visually outline meaningful structural information, such as orthant space usage and the geometric distribution of data points and classes. Its hierarchical, variance-ordered concentric ring structure effectively translates the complexity of $\mathbb{R}^d$ space into an intuitive format that helps in the identification of clusters and sparse regions.

Although RAYS has inherent limitations, particularly its reduced efficacy when dealing with extremely uniform, high-dimensional datasets, it provides crucial information to data analysts in an easily understandable method. By providing a clear and comprehensive representation of the complete dataset in a single view, RAYS enables the naked eye to grasp the fundamental spread, sparsity, and class commonality of the dataset at a glance.

7 AI Use Statement

Generative AI tools were used to assist with grammar checking, sentence refinement, and improving clarity of technical explanations in this manuscript. All scientific content, methodology, analysis, and conclusions are the work of the author. The AI was not used to generate scientific claims or interpret results.

References

1. Johansson Westberg, Jimmy Forsell, Camilla. (2015). Evaluation of Parallel Coordinates: Overview, Categorization and Guidelines for Future Research. IEEE Transactions on Visualization and Computer Graphics. 22. 1-1. 10.1109/TVCG.2015.2466992.
2. Ankerst, Mihael Keim, Daniel Kröger, Peer. (1996). Circle Segments : A Technique for Visually Exploring Large Multidimensional Data Sets. First publ. in: Visualization '96, Hot Topic Session, San Francisco, CA, November, 1996.
3. Hemant Makwana, Sanjay Tanwani, Suresh Jain . Axes Re-Ordering in Parallel Coordinate for Pattern Optimization. International Journal of Computer Applications. 40, 13 (February 2012), 43-48. DOI=10.5120/5044-7370
4. Jäckle, Dominik Hund, Michael Behrisch, Michael Keim, Daniel Schreck, Tobias. (2017). Pattern Trails: Visual Analysis of Pattern Transitions in Subspaces. 1-12. 10.1109/VAST.2017.8585613.